



IDP

Slingshot Launcher

TEAM 101

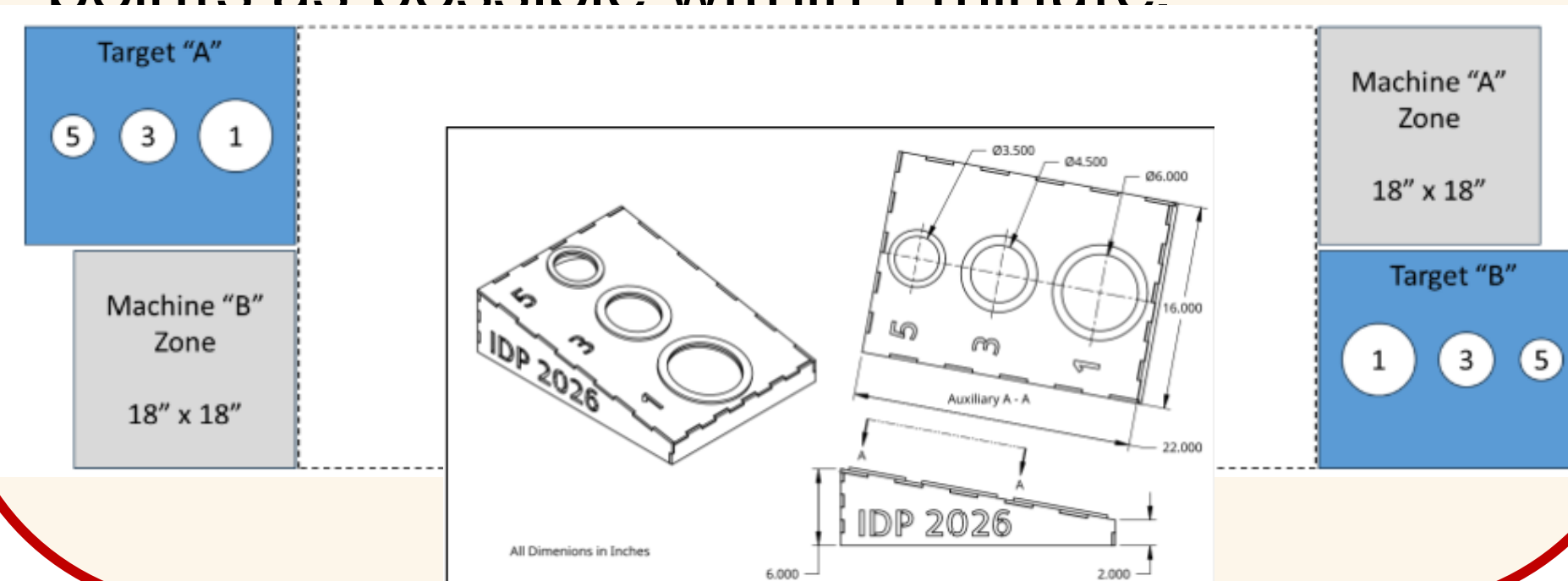
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DESIGN PROBLEM

Design Challenge: Design and fabricate a purely mechanical launcher to score points by launching bean bags into a cornhole-style target positioned 10 feet away from the launcher.

Constraints: Fit within 18" X 18" area, use materials from given list, purely mechanical, no electronics. Score as many points as possible within 1 minute.



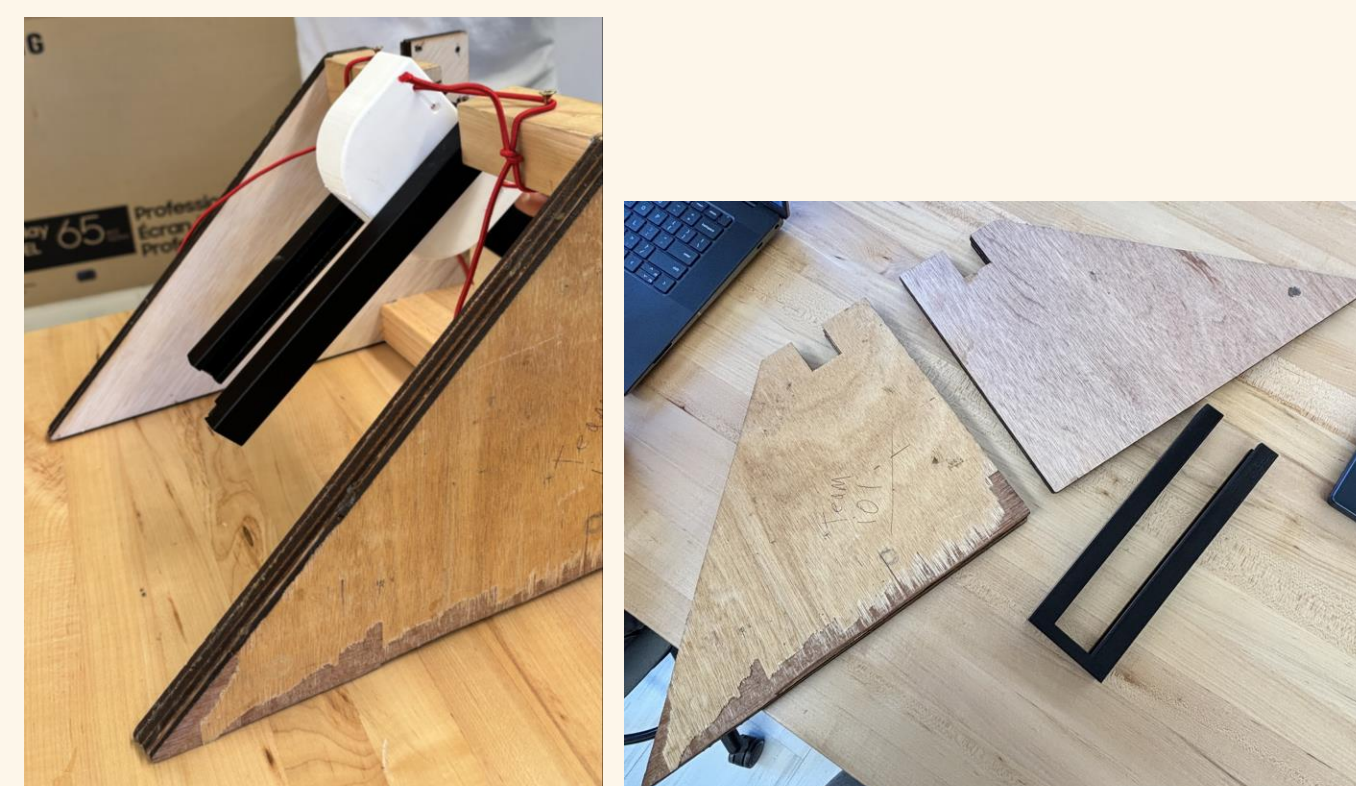
PROJECT BACKGROUND

Bungee-powered launchers are a widely used approach. These systems **store energy by stretching elastic cords**, which can then be released to propel an object forward.

Elastic launchers are mechanically simpler and allow easier adjustment of launch conditions. The amount of stored energy can be controlled by changing the pullback distance of the elastic cords.



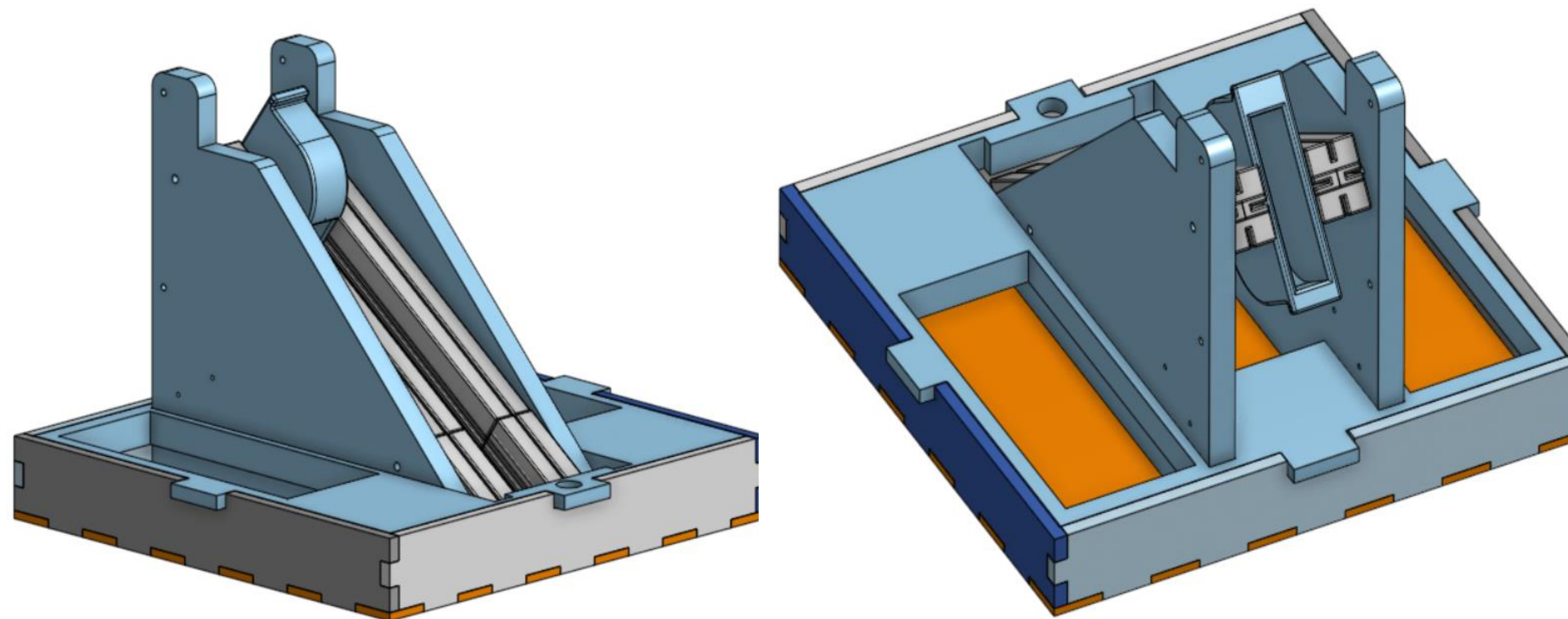
INITIAL DESIGN



The triangular frame was selected to provide stability and support the forces generated during operation, while the sliding basket was designed to maintain a **consistent launch path**.

During the design process, adjustability was an important consideration. **The pullback distance could be varied to change the launch power**, allowing the team to test different settings and evaluate launcher performance.

DESIGN SOLUTION



ELASTIC ENERGY COMPONENT

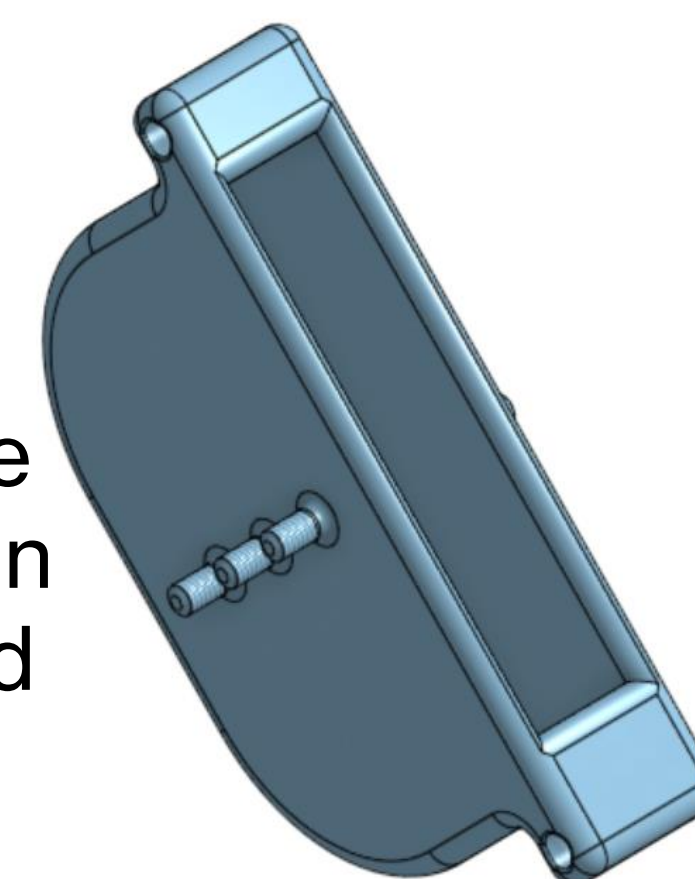
Bungee cord stores elastic potential energy when stretched and releases it as kinetic energy **to launch the bean bag**. Selected for its adjustable and repeatable launch force.



BASKET DESIGN

Material: PLA

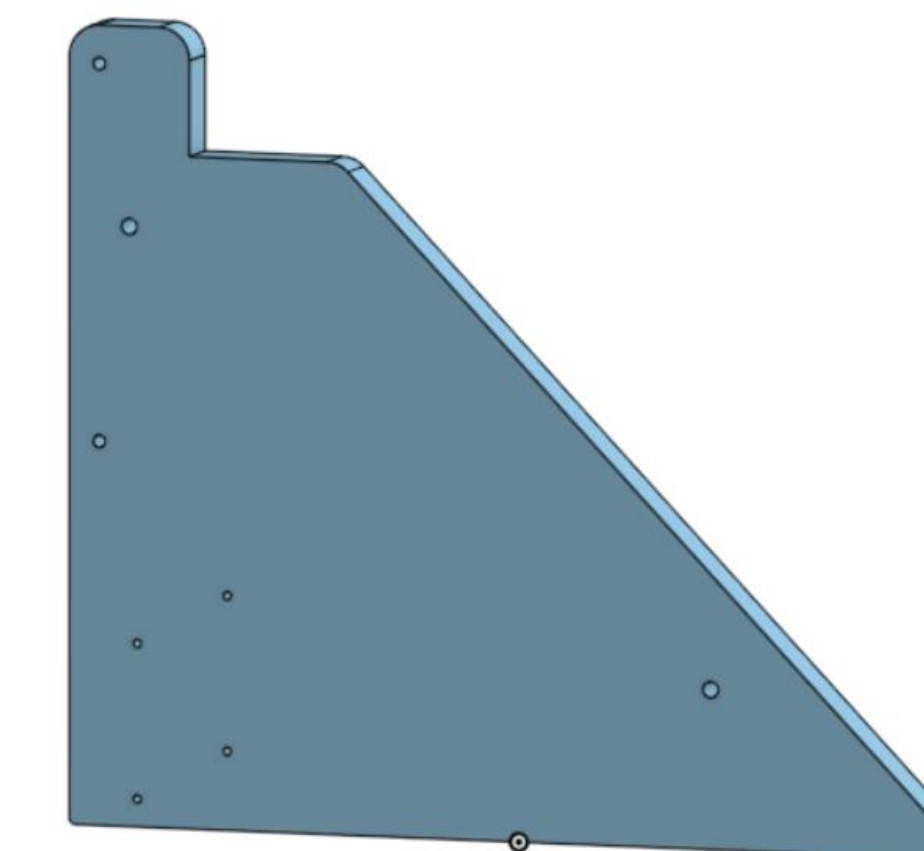
The design we chose directly slide into the slider rail. With minimal friction. The basket will hold a single beanbag at a time with an opening in back so the bag can be easily pulled in and oriented in an **optimal launching position**.



FRAME DESIGN

Material: Laser Cut Plywood.

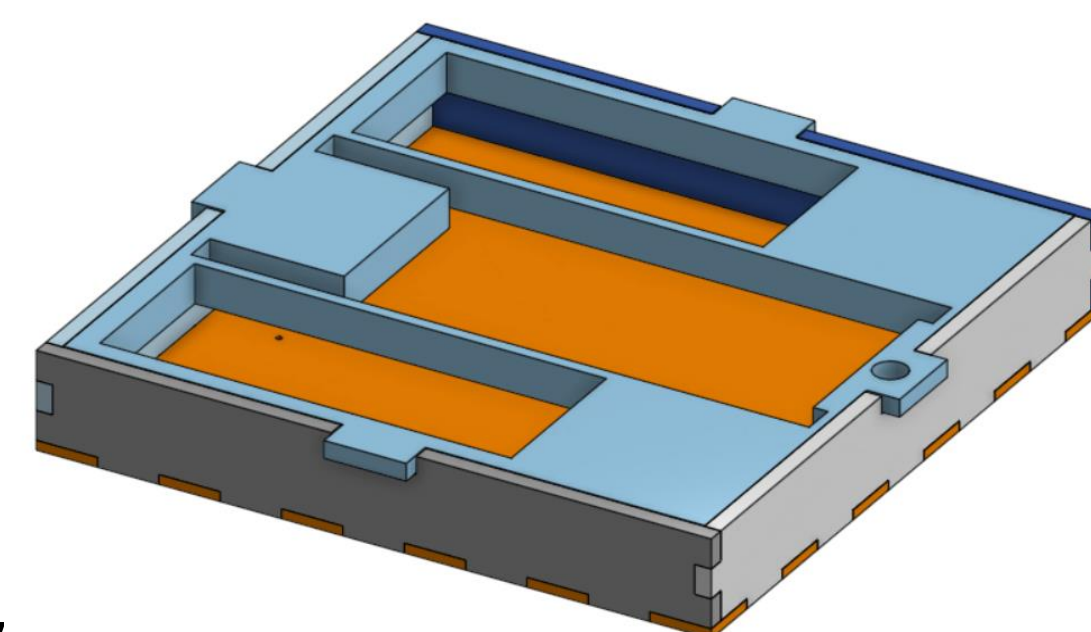
Triangle design for the frame to optimize the **most stability** and to keep our slider fixed to a 35-degree angle.



BASE DESIGN

Material: 1/4 in Plywood.

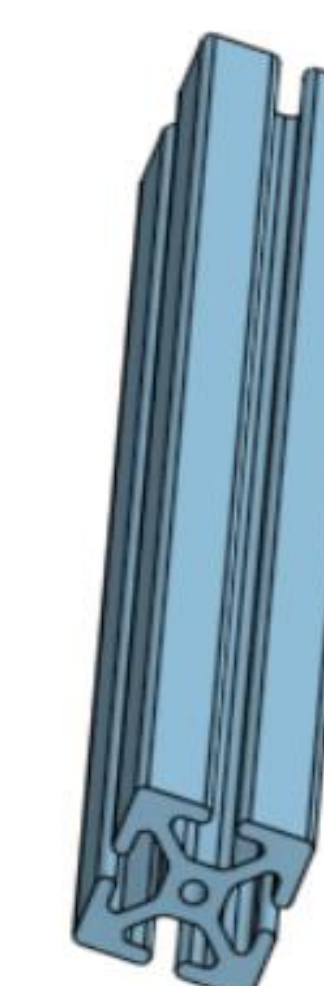
Laser-cut plywood provides a lightweight, rigid base that supports the launcher structure, improves stability, and **allows precise assembly of all components**.



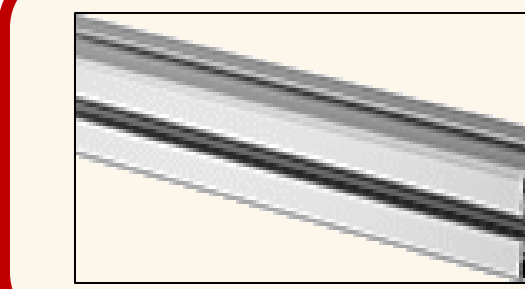
SLIDER RAILS DESIGN

Material: Aluminum Extrusion.

Provides a strong, durable, and low-friction guide system for the basket, allowing smooth and consistent sliding during launch.

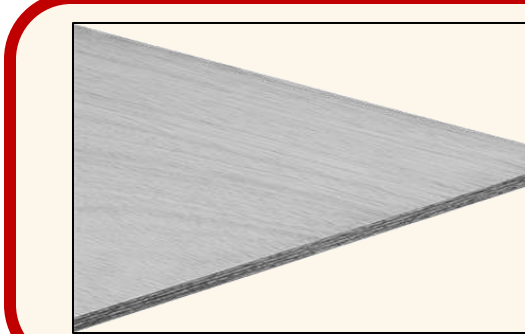


COMPONENTS



1. Extrusion

Material: s Aluminum
Cost: \$10.67
Purpose: guide the bean bag through the launch



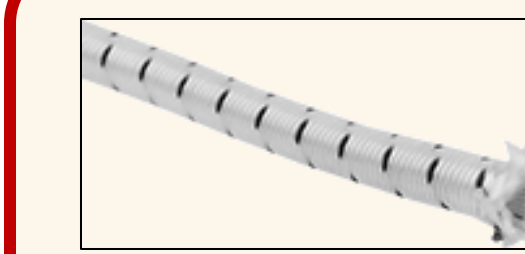
2. Plywood

Material: ply-wood
Cost: \$27
Purpose: stabilize the launch for the beanbag



3. Wood

Material: 2x4
Cost: \$2.25
Purpose: Help build base and keep it stable



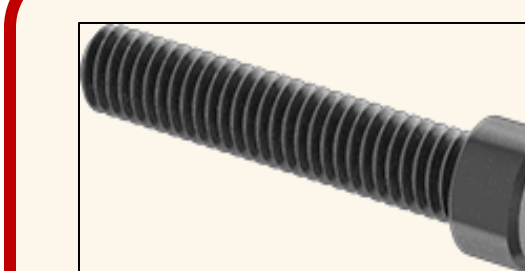
4. Bungee Cord

Material: Rubber band
Cost: \$0.80
Purpose: Help build tension to release the bean bag



5. 3D Printed Nut

Material: Steel drop T-Nut
Cost: \$0.30
Purpose: Help hold together steel extrusion and plywood



6. Screw

Material: 2 1/4-20x1-1/4 screw
Cost: \$1.00
Purpose: Help keep everything together.

Total Cost: \$42.02

FUTURE WORK

Additional improvements may be achieved by exploring different **material selections for critical components**. Adopting a stronger, lower-friction basket material to improve launch consistency, durability, and operational efficiency.

TARGET MARKET

The proposed launcher has potential applications in both recreational and educational markets.

It could provide a fun and interactive alternative to traditional cornhole for backyard games and competitions, while also **serving as a hands-on tool for demonstrating engineering concepts** such as projectile motion, energy storage, and design optimization.